

Hasan Shaikh

📍 Christian Medical College, Vellore, Tamil Nadu - 632002, India

☎ +91-7906049358 | ✉ hasanshaikh3198@gmail.com | [in LinkedIn](#) | [GitHub](#) | [Website](#) | [Scholar](#) | [ORCID](#)

📁 Work Experience

Quantitative Imaging Research and Artificial Intelligence Lab (QIRAIL [↗](#))
Project Assistant, Christian Medical College Vellore, Dept. of Radiation Oncology

Tamil Nadu, India
Aug. 2024 – Present

[P1] Large-Scale Prospective Imaging Study – DBT/Wellcome Trust India Alliance

- Contributed to the end-to-end radiomics pipeline for **1,900+ head and neck cancer patients** in a DBT/Wellcome Trust-funded prospective study (2020-2025) — DICOM retrieval, automated tumor segmentation, and PyRadiomics feature extraction.
- Deployed AWS S3 backup pipelines with error recovery, and helped coordinate annotation workflows across radiation oncology sites.

[P2] Radiomics-Based Risk Stratification in Head and Neck Cancer

- Developed a 10-feature prognostic signature (2 clinical + 8 radiomics) achieving **AUC 0.82 (95% CI: 0.62–0.95)** for locoregional recurrence prediction in 163 patients, selected from 1000+ candidate features.
- Benchmarked 5 metaheuristic feature selection algorithms — Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), Whale Optimization Algorithm (WOA), and Simulated Annealing (SA) — across multiple classifiers to find which works best for high-dimensional radiomics data.

[P3] Automated Segmentation of Head and Neck Cancer Using 3D CNNs – Collaboration with NIT Surathkal

- Curated a multi-institutional dataset (167 cases: 137 MAASTRO + 30 CMC) and found a **7% performance drop** on Indian patients relative to European data.
- Trained **3D nnU-Net** achieving **Dice: 0.62 (MAASTRO), 0.63 (CMC), 0.65 (combined)** for tumor segmentation from CT scans alone.

[P4] CHAVI: CompreHensive Digital ArchiVe of Cancer Imaging – India's First National Oncology Biobank

- Contributed **304+ de-identified cases** with validated clinical and imaging metadata to India's first national oncology imaging biobank (collaboration with Tata Memorial Centre and IIT Kharagpur), using DICOM and XNAT pipelines.

[P5] Reproducibility Study: CNN-Based Head and Neck Cancer Prognosis [↻](#)

- Audited a published CNN model for HNC prognosis across three outcomes (distant metastasis, locoregional failure, overall survival); identified incomplete datasets, undocumented preprocessing, and broken validation protocols.
- Corrected all identified errors and successfully reproduced the results; the original authors formally acknowledged the issues, contributing to improved reproducibility standards in published medical AI.

STARlab Capital
Research Analyst

Lucknow, India
Dec. 2023 – June 2024

- Designed and backtested volatility-based options strategies (Nebula, ARUT, A2) using **OptionNet Explorer**, **Mesosim**, **OptiTrade**, **OptiBot** tools.
- Led contributions to **OptiTrade's open-source repository**, improving internal backtesting and trade log systems.

Publications & Research

- [1] **Hasan Shaikh**, Sathya A, Praveenraj C, Balu Krishna S, Aparna Irodi, Rajendra Benny Kuchipudi, Jino Wilson Victor, Manu Mathew, Rajesh Isiah, Simon Pavamani, Joy Mammen, Hannah Mary Thomas T, “*Can reproducibility be achieved? External validation of ‘Reproducible’ image-based prognostic models for head and neck cancer*”, *Scientific Reports, Nature Portfolio*. [Under Submission]
- [2] Balu Krishna S, Amal Joseph Varghese, **Hasan Shaikh** et al., “*Development and validation of a prospective radiomics-clinical signature for locoregional recurrence in patients with locally advanced head and neck cancer*”, *European Society for Radiotherapy and Oncology (ESTRO)*, Stockholm, Sweden, 2026. [Abstract] [Poster]
- [3] **Hasan Shaikh**, Balu Krishna S, Amal Joseph Varghese, et al., “*Metaheuristic-Driven Machine Learning Pipelines for Radiomics-Based Prediction of Locoregional Recurrence in Head and Neck Cancer*”, In: *Proceedings of the International Conference on Artificial Intelligence for Healthcare (AIHC), Lecture Notes in Electrical Engineering, Springer Nature*, 2025. [Accepted]
- [4] Piyus Prabhanjans, Asjad Nabeel P, Aparna V K, Hannah Mary Thomas T, Balu Krishna Sasidharan, **Hasan Shaikh**, Amal Joseph Varghese, Rajendra Benny Kuchipudi, Simon Pavamani, Jeny Rajan, “*Automated Segmentation of Head and Neck Cancer from CT Images Using 3D Convolutional Neural Networks*”, *medRxiv* (preprint), 2026. <https://doi.org/10.64898/2026.03.12.26347996>
- [5] **Hasan Shaikh** and Rashid Ali, “*Cancer Survival Prediction Using Artificial Intelligence: Current Status and Future Prospects*”, *Data Science in the Medical Field, Academic Press, Elsevier*, 2024. <https://doi.org/10.1016/B978-0-443-24028-7.00016-7> [Book Chapter]
- [6] **Hasan Shaikh**, Amal Joseph Varghese, Balu Krishna S, et al., “*Before We Treat, Can We Tell? A Locoregional Recurrence Signature in Head & Neck*”, *15th Research Day*, Christian Medical College, Vellore, India, 2025. [Poster]
- [7] **Hasan Shaikh**, Amal Joseph Varghese, Hannah Mary Thomas T et al., “*Can CT Radiomics Predict Recurrence in Head and Neck Cancer? Early Results from a Prospective Imaging Trial*”, *14th Research Day*, Christian Medical College, Vellore, India, 2024. [Poster]

Master's Thesis

Hasan Shaikh, “*Multimodal Data Analytics for Predicting the Survival of Cancer Patients*”, Aligarh Muslim University, 2023.
Advisor: Prof. Rashid Ali. 

- Implemented **MGAttCNNMD** (Multimodal Gated Attention Convolutional Neural Network integrating Multidimensional Data) — a deep learning architecture for breast cancer survival prediction on the METABRIC dataset.
- Built three Gated Attention CNNs, one each for clinical records, genomic copy number alterations, and gene expression profiles, fused through a stacked Random Forest — achieving **91.2% survival prediction accuracy**.

Technical Skills

Machine Learning & DL: TensorFlow, Keras, scikit-learn, PyRadiomics, nnU-Net

Programming Language : Python (NumPy, Pandas, SciPy), Bash

Tools : 3D Slicer, Orthanc PACS, XNAT, DICOM, NIfTI, Docker, Github

Specialized Methods : Radiomics, Metaheuristic Techniques, Graph Neural Networks

Education

Master of Technology in Computer Science & Engineering

Nov. 2021 – Nov. 2023

Aligarh Muslim University (AMU), CGPA: 8.80 / 10.00

Aligarh, India

Bachelor of Technology in Computer Science and Engineering

Aug. 2017 – Jul. 2021

Dr. A.P.J. Abdul Kalam Technical University (AKTU), CGPA: 8.04 / 10.00

Lucknow, India

Academic Achievements & Honors

- **Third Prize, Oral Presentation** at the 2nd National Symposium on Health Data and AI (March 2025) — Presented “*Automated Segmentation of Head and Neck Cancer from CT Images Using 3D nnU-Net*”, awarded by the BioMedical Informatics Unit, CMC Vellore.

- **Honors Degree** — Ranked in the top 1% out of 128 students in the undergraduate engineering program.
- **NPTEL Courses conducted by IITs (Elite + Silver Certified):**
 - Data Analytics with Python (80%) ([Link](#))
 - Essential Mathematics for Machine Learning
 - Deep Learning
 - Python for Data Science (78%) ([Link](#))
 - Machine Learning
 - Demystifying the Brain ([Link](#))

Relevant Courses

Relevant courses completed during my Bachelor's and Master's programs include:

- Artificial Intelligence
- Machine Learning
- Web Mining & Searching
- Big Data Analytics
- Image Processing & Computer Vision
- Application of Soft Computing
- Data Structures
- Applied Linear Algebra
- Design & Analysis of Algorithms
- Database Management Systems
- Data Warehousing & Data Mining
- Distributed Systems

Conferences & Workshops

1. Presented paper at the **International Conference on Artificial Intelligence for Healthcare (AIHC)**, organized by National Institute of Technology (NIT) Calicut, India (December 10 - 12, 2025) on Metaheuristic-driven radiomics-based locoregional recurrence prediction in head and neck cancer. [*Lecture Notes in Electrical Engineering, Springer*] ([Link](#))
2. Organized the **1st & 2nd Workshop on Radiomics and Auto Segmentation**, conducted by the Department of Radiation Oncology & Quantitative Imaging Research and Artificial Intelligence Lab, Christian Medical College (CMC) Vellore, Tamil Nadu, India (November 14–15, 2025 & March 27-28, 2026). ([Link](#))
3. Organized the **Head and Neck Cancer Survivorship Program**, conducted by the Department of Radiation Oncology, Christian Medical College (CMC) Vellore, Tamil Nadu, India (September 12–13, 2025). Themes: "Together We Overcome, Together We Heal" and "De-Escalation Strategies in Head and Neck Cancer."
4. Participated and served as part of the Organizing Team for the **2nd Annual Winter Symposium on Health Data and AI**, conducted by Biomedical Informatics Unit, Christian Medical College (CMC) Vellore, Tamil Nadu, India (March 13–15, 2025).
5. Attended the Continuing Medical Education (CME) program on **Revolution and Precision in Radiation Oncology**, Ida B. Scudder Cancer Center, Christian Medical College (CMC) Vellore, Tamil Nadu, India (March 1, 2025).
6. Participated in the **14th & 15th Annual Research Day**, organized by the Office of Research, Christian Medical College (CMC) Vellore, Tamil Nadu, India (October 24–25, 2024 & 2025).
7. **AI & ML for Engineering & Social Sciences Research**, 2023 IEEE Computational Intelligence Society (CIS) Summer School, organized by Malaviya National Institute of Technology Jaipur (MNIT) Jaipur, India, 4 – 8 Sep. 2023. ([Link](#))
8. **7th Summer School on AI with Focus on Computer Vision & ML**, coordinated by International Institute of Information Technology (IIIT) Hyderabad, India, 1 – 31 Aug. 2023. ([Link](#))
9. **Emerging Research Trends in Computational Intelligence Techniques to Address Challenges in Biomedical Data and Imaging**, 2022 IEEE CIS Summer School, organized by National Institute of Technology (NIT) Arunachal Pradesh, India, 7 – 11 Nov. 2022. ([Link](#))

Referees

Dr. Hannah Mary Thomas T
 Scientist, Biomedical Informatics Unit
 India Alliance Early Career Fellow
 Christian Medical College, Vellore, India
 Phone: +91-96262 62292
hannah.thomas@cmcvellore.ac.in

Dr. Jeny Rajan
 Associate Professor, Department of Computer Engineering
 National Institute of Technology Karnataka (NITK)
 Surathkal, Mangalore, India
 Phone: +91-7907595704
jenyrajan@nitk.edu.in